

GOSFORD HIGH SCHOOL



Year 11

MATHEMATICS EXTENSION 1

Preliminary Course

Assessment Task 1

2010

Time Allowed: 70 minutes

Instructions:

- There are 5 questions.
- Answer each question on your own paper showing all necessary working.
- Start **EACH** question on a new sheet of paper.
- Calculators may be used.

Question	Mark
1. Arithmetic / Algebra	/10
2. Polynomials	/10
3. Counting Theory	/10
4. Inequalities	/10
5. Further Algebra	/10
TOTAL	/50

QUESTION 1: (10 Marks)

- | | | Marks |
|----|--|-------|
| a. | Calculate: | 2 |
| | $\frac{76.5}{17 \times (81.76)^3} \times \frac{1851.7 - 24^2}{\sqrt{47.8}}$ | |
| | writing your answer in scientific notation, correct to 3 significant figures. | |
| b. | A builder intends to pay cash for building materials and receives a discount of $12 \frac{1}{2} \%$ for doing so. As well, he receives a further 10% trade discount on this cash price. If he eventually pays \$1055, find the original price of the building materials. (Answer to the nearest dollar.) | 2 |
| c. | Evaluate $\left(\frac{2^x + 2^x + 2^x + 2^x}{2^{x+1}} \right)^2$ | 2 |
| d. | If $\frac{a^2 b^2}{c^4} \times \frac{c^3}{a^5 b^3} \times \frac{(a^2 c)^3}{b^2} = a^x b^y c^z$,
find the value of x, y, z | 2 |
| e. | If $(2\sqrt{3} - \sqrt{7})^3 = m\sqrt{3} + n\sqrt{7}$,
find the value of m, n | 2 |

QUESTION 2: (10 Marks)

	Marks
a. For the polynomial $P(x) = 2x^4 - x^2 + x - 3$ State the i. Degree ii. Constant term	1
b. Which of the expressions: A. $5 - x^2 - x^3$ B. $7x^2 + \frac{7}{x^2}$ C. 3 D. $x\sqrt{x} + \sqrt{x}$ are polynomials?	1
c. What is a monic polynomial? Explain and give an example.	1
d. Divide $P(x) = x^4 + x^3 - x^2 + x - 2$ by $x^2 - 1$ and express your answer in the form $P(x) = A(x)Q(x) + R(x)$	2
e. Find the zeros of the polynomial $Q(x) = x^4 - 2x^3 - 8x^2$ and use them to sketch the graph of $y = Q(x)$ (Indicate all intercepts on your sketch)	2
f. Show that $x - 1$ is not a factor of $G(x) = x^5 - 2x^4 + 7x^2 - 3x + 5$	1
g. When $F(x) = ax^2 - 3x + 1$ and $H(x) = x^3 - 3x + 2$ are divided by $x + 1$ they have the same remainder. Find the value of a.	2

QUESTION 3: (10 Marks)

	Marks
a. Show how you would calculate 6P_4 . (Do not evaluate)	1
b. Two different numbers are chosen at random from 1, 2, 3, 4, 5	1
i. How many different combinations can be chosen?	1
ii. How many of these combinations have an even sum when the two numbers are added together?	1
c. Let each different arrangement of all the letters of INTERESTED be called a word. How many words are possible?	1
d. In how many ways can 6 girls sit at a round table if:	1
i. There are no restrictions	1
ii. 3 particular girls insist on sitting together	1
e. A ticket queue has 5 men and 5 women standing in a line. Find how many different arrangements of the line are possible if	
i. the men and women can stand anywhere in the line	1
ii. the men and women alternate	1
iii. 2 particular women cannot stand together.	2

QUESTION 4: (10 Marks)

	Marks
a. Solve $\frac{4}{5-x} \leq 1$	2
b. Solve $\frac{x^2-4}{x} > 0$	2
c. Solve $\frac{5x+9}{ x-5 } > 1$	3
d. Solve and graph the solution(s) to: $ x+4 + x-4 = 8$	3

QUESTION 5: (10 Marks)

	Marks
a. Express as a single fraction: $(x^{-1} + y^{-1})^{-1}$	2
b. If $x = \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$, express $x - \frac{1}{x}$ in simplest form with a rational denominator.	3
c. Simplify: $\frac{20x + 12y - 25x^2 + 9y^2}{25x^2 - 9y^2} \div \frac{8 - 10x + 6y}{25x^2 - 30xy + 9y^2}$	3
d. i. Express $x^4 + 4$ in the form $(x^2 + a)^2 - (bx)^2$ and find the value of a and b	1
ii. Hence, express $x^4 + 4$ as a product of two quadratic factors	1

END OF TEST

Q1/

a) $0.001519233 \dots$ (calc.)
 $= 1.52 \times 10^{-3}$

b) Amount paid:
 90% of $87\frac{1}{2}\%$
 $= 78\frac{3}{4}\%$
 $\therefore 78\frac{3}{4}\% = \1055
 $1\% = \$13.397 \dots$
 $100\% = \$1339.7 \dots$

$\therefore$ Original Price = \$1340

c) $\left(\frac{2^x + 2^x + 2^x + 2^x}{2^{x+1}}\right)^2$
 $= \left(\frac{4 \cdot 2^x}{2^{x+1}}\right)^2$
 $= \left(\frac{2^2 \cdot 2^x}{2^{x+1}}\right)^2$
 $= \left(\frac{2^{x+2}}{2^{x+1}}\right)^2$
 $= 2^2$
 $= 4$

d) $\frac{a^2 b^2}{c^4} \times \frac{c^3}{a^5 b^3} \times \frac{(a^3 c)^3}{b^2}$
 $= \frac{a^2 b^2 c^3}{a^5 b^3 c^4} \times \frac{a^6 c^3}{b^2}$
 $= \frac{a^8 b^2 c^6}{a^5 b^5 c^4} = \frac{a^3 c^2}{b^3}$

$\therefore x = 3, y = -3, z = 2$

e) $(2\sqrt{3} - \sqrt{7})^3$
 $= (2\sqrt{3} - \sqrt{7})(2\sqrt{3} - \sqrt{7})^2$
 $= (12 - 4\sqrt{21} + 7)(2\sqrt{3} - \sqrt{7})$
 $= (19 - 4\sqrt{21})(2\sqrt{3} - \sqrt{7})$
 $= 38\sqrt{3} - 19\sqrt{7} - 8\sqrt{63} + 4\sqrt{147}$
 $= 38\sqrt{3} - 19\sqrt{7} - 24\sqrt{7} + 28\sqrt{3}$
 $= 66\sqrt{3} - 43\sqrt{7}$
 $\therefore m = 66, n = -43$

Q2/

a) (i) Degree = 4 $\left(\frac{1}{2}\right)$
 (ii) Constant = -3 $\left(\frac{1}{2}\right)$

b) A and C (i)

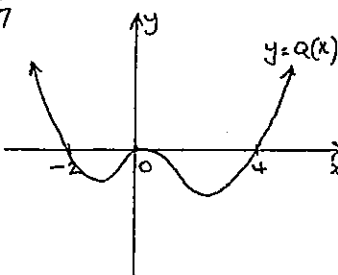
c) A monic polynomial is a polynomial with the leading co-efficient equal to 1. $\left(\frac{1}{2}\right)$

eg: $x^3 - 4x^2 + 5x - 7$ $\left(\frac{1}{2}\right)$
 (of course others)

d) $\frac{x^2 + x}{x^2 - 1} \div \frac{x^4 + x^3 - x^2 + x - 2}{x^4 - x^2}$
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^4 - x^2}{x^4 + x^3 - x^2 + x - 2}$
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^2(x^2 - 1)}{x^2(x^2 + x - 2)}$
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^2(x^2 - 1)}{x^2(x^2 + x - 2)}$
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^2(x^2 - 1)}{x^2(x^2 + x - 2)}$

$\therefore P(x) = (x^2 - 1)(x^2 + x) + (2x - 2)$

e) $Q(x) = 0$ for zeros
 $\therefore x^2(x^2 - 2x - 8) = 0$
 $x^2(x - 4)(x + 2) = 0$
 $\therefore$ zeros = 0, 4, -2



f) $G(1) = 1 - 2 + 7 - 3 + 5 = 8$

Since $G(1) \neq 0$
 $x - 1$ is not a factor

g) $F(-1) = a + 3 + 1 = a + 4$

$H(-1) = -1 + 3 + 2 = 4$

If the same remainder:
 $\therefore a + 4 = 4$
 $a = 0$

Q3/

a) Using ${}^n P_r = \frac{n!}{(n-r)!}$
 ${}^6 P_4 = \frac{6!}{(6-4)!} = \frac{6!}{2!}$
 $= 6 \times 5 \times 4 \times 3$

Q3/ b) (i) ${}^5 C_2 = 10$
 (ii) either 2 odds or 2 evens
 $\therefore (2+4), (1+3), (3+5)$
 (1+5)

Answer = 4

c) $\frac{10!}{2!3!} = 302400$

d) (i) using $(n-1)!$
 $= 5!$

$= 120$ ways.

(ii) Grouping 3 girls together in a circle
 $= (4-1)!$
 $= 3!$ and

the 3 girls can be placed in $3!$ ways
 total ways = $3! \times 3!$
 $= 36$

(e) (i) $10! = 3628800$
 (ii) Men $\rightarrow 5!$
 Women $\rightarrow 5!$

Now either
 mwmwmwmwmw
 or
 wmwmmwmwmw

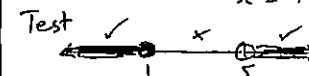
$\therefore$ Ways = $2 \times 5! \times 5!$
 $= 28800$

(iii) 2 particular women together = $2 \times 9!$

$\therefore$ not together
 $= 10! - 2 \times 9!$
 $(3628800 - 806400)$
 $= 2822400$

Q4/ a) Consider $\frac{4}{5-x} = 1$

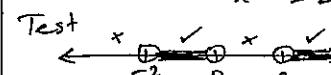
C.V: $x \neq 5$
 $4 = 5 - x$
 $x = 1$



$\therefore x \leq 1, x > 5$

b) Consider $\frac{x^2 - 4}{x} = 0$

C.V: $x \neq 0$
 $x^2 - 4 = 0$
 $x = \pm 2$



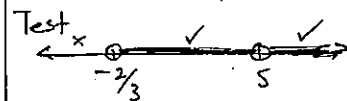
$\therefore -2 < x < 0, x > 2$

c) Consider $\frac{5x+9}{|x-5|} = 1$

$x \neq 5$
 $5x+9 = |x-5|$

$5x+9 = x-5$ or $5x+9 = -x+5$
 $4x = -14$ $6x = -4$
 $x = -\frac{7}{2}$ $x = -\frac{2}{3}$

check $x = -\frac{2}{3}$ is only soln to equation



$-\frac{2}{3} < x < 5, x > 5$

d) $|x+4| + |x-4| = 8$
 4 cases: 1) $x+4 > 0$ and $x-4 > 0$

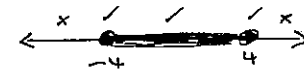
$x+4 + x-4 = 8$
 $2x = 8$
 $x = 4$ ✓

2) $x+4 < 0$ and $x-4 < 0$
 $-(x+4) - (x-4) = 8$
 $-x-4 - x+4 = 8$
 $-2x = 8$
 $x = -4$ ✓

3) $x+4 > 0$ and $x-4 < 0$
 $(x+4) - (x-4) = 8$
 $x+4 - x+4 = 8$
 $x+4 - x+4 = 8$
 $8 = 8$

i.e. in this region $-4 < x < 4$ equation is always true

4) $x+4 < 0$ and $x-4 > 0$
 $-(x+4) + (x-4) = 8$
 $-x-4 + x-4 = 8$
 $-8 = 8$
 $\therefore$ no solutions for $x < -4$ and $x > 4$



4/ a) $(x^{-1} + y^{-1})^{-1}$
 $= \left(\frac{1}{x} + \frac{1}{y}\right)^{-1}$
 $= \left(\frac{y+x}{xy}\right)^{-1}$
 $= \frac{xy}{x+y}$

Q.5 b)

$$x - \frac{1}{x} = \frac{3-2\sqrt{2}}{3+2\sqrt{2}} - \frac{3+2\sqrt{2}}{3-2\sqrt{2}}$$

$$= \frac{(3-2\sqrt{2})^2 - (3+2\sqrt{2})^2}{(3+2\sqrt{2})(3-2\sqrt{2})}$$

$$= \frac{9 - 12\sqrt{2} + 8 - (9 + 12\sqrt{2} + 8)}{9 - 8}$$

$$= \frac{17 - 12\sqrt{2} - 17 - 12\sqrt{2}}{1}$$

$$= -24\sqrt{2}$$

$$c) \frac{20x + 12y - 25x^2 + 9y^2}{25x^2 - 9y^2} \times \frac{25x^2 - 30xy + 9y^2}{8 - 10x + 6y}$$

$$= \frac{4(5x+3y) - (5x-3y)(5x+3y)}{(5x-3y)(5x+3y)} + \frac{(5x-3y)(5x-3y)}{2(4-5x+3y)}$$

$$= \frac{(5x+3y)(4-5x+3y)}{(5x-3y)(5x+3y)} \times \frac{(5x-3y)(5x-3y)}{2(4-5x+3y)}$$

$$= \frac{5x-3y}{2}$$

$$d) (i) x^4 + 4 = x^4 + 4x^2 + 4 - 4x^2 \\ = (x^2 + 2)^2 - (2x)^2$$

$$\therefore a = 2, b = 2$$

$$(ii) x^4 + 4 =$$

$$\left[(x^2 + 2) - 2x \right] \left[(x^2 + 2) + 2x \right]$$

$$= (x^2 - 2x + 2)(x^2 + 2x + 2)$$